CSE440: Natural Language Processing II

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Lecture 3: ML Essentials

Topics

- Classification
- Feature representation
- Probability
- Naive Bayes
- ML evaluation

Classification

- Let’s see an example from “*Where’s My Mom?* by Julia Donaldson and Axel Scheffler ISBN-13: 978-0803732285”
- Context: a baby monkey is looking for his mom, and cannot find her anywhere
- A butterfly comes to help, asks the baby monkey how his mother looks like
- And the story starts

Who is the classifier here? What are the features? What is a feature?

Classifier

- Classifier = butterfly
- Features
  - bigger than baby
  - not a great gray hunk
  - no tusks, trunk
  - knees not baggy
  - tail coils around trees
  - doesn’t slither, hiss
  - no nest of eggs
  - legs > 0

Classifier

- Objects are described by properties or features
- Classifiers make predictions based on properties
- Good classifiers consider many properties jointly
- Classifiers may overfit, i.e., perform well on training data, but poorly on unseen data

Two types of classifier:

1. **Generative Classifier:** Models how input data is generated
2. **Discriminative Classifier:** Models the decision boundary between classes

We will mostly use discriminative classifier in this course– where the classifier function tries to draw (or predicts) a border to separate different types of data.

Classifier

- A classifier needs to be trained
- We train a classifier on a set of data we call training data
- We try to *fit* the classifier as best to our ability to the training data
    - Should we do this? Will learn soon
- Then we try to predict the class of a new data that is not seen during training
    - This is the part where prediction comes in

Formal definitions

A classifier, $h \in O \longrightarrow Y$, is a function that maps an input object, $o \in O$ to an output category, $y \in Y$.

- Example: a student retention classifier, where
  $O = \mathit{all\ UA\ students}$ and
  $Y = \{\mathit{will\text{-}graduate}, \mathit{won't\text{-}graduate}\}$

A feature function, f_i, maps an input object to a (numeric) feature value representing some property of the object.

- Example: GPA feature, $f_{GPA}(\mathit{Steven\ Bethard}) = 4.0$

A feature vector is a representation of input data, where each element of the vector is the value of one feature:

$$x = [f_1(o), f_2(o), \ldots, f_m(o)]$$

Classification in matrix form

Most classification algorithms are trained on a large sample of input objects, $\hat{O}$, whose output categories are known:

$$\textit{training data} = \{(o_i, y_i) : o_i \in \hat{O}, y_i \in Y\}$$

If we have m feature functions, then we usually view this dataset as an $\mathbf{X}$ matrix and a Y vector:

$$\overbrace{\begin{bmatrix} f_1(o_1) & f_2(o_1) & f_3(o_1) & \cdots & f_m(o_1) \\ f_1(o_2) & f_2(o_2) & f_3(o_2) & \cdots & f_m(o_2) \\ f_1(o_3) & f_2(o_3) & f_3(o_3) & \cdots & f_m(o_3) \\ \vdots & \vdots & \ddots & \vdots & \\ f_1(o_n) & f_2(o_n) & f_3(o_n) & \cdots & f_m(o_n) \end{bmatrix}}^{\mathbf{X}} \quad \overbrace{\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix}}^{Y}$$

Classifier training will try to construct h such that $h(\mathbf{X}) = Y$.

Features are easy for regular ML task

- Like you have seen in the butterfly classifier, or other examples
- What features do we use when we are trying to classify a natural language data?
- Let’s see an example: BoW features

Bag-of-words features

- A bag-of-words feature representation means that for each word in the vocabulary, there is a feature function, f_i that produces the count of word i in the text.
- Example: f_{great} (great scenes great film) = 2

Documents	f_{and}	f_{boxing}	f_{film}	f_{great}	f_{plot}	f_{satire}	f_{scenes}	f_{twists}	f_{worst}
worst boxing scenes	0	1	0	0	0	0	1	0	1
satire and great plot twists	1	0	0	1	1	1	0	1	0
great scenes great film	0	0	1	2	0	0	1	0	0

Bag-of-words features

How do we do it?

- Very easy
- Create a set of all possible unique words in the train data, w
- For each of the sentences create a vector v of size |w| with every element initialized to 0
- If a word i appears in that sentence, replace the 0 in v_i with the count of that word in that sentence
- That’s it

What are the issues with BoW?

- One feature function per word → large, sparse matrices
  - What’s wrong with sparsity?
- Completely ignores word order
- But often still useful

Classwork

Construct the feature matrix for these four text messages:

- Sorry I’ll call later
- U can call me now
- U have won call now!!
- Sorry! U can not unsubscribe

Classwork

	$f_{!}$	$f_{'ll}$	f_{I}	f_{Sorry}	f_{U}	f_{call}	f_{can}	f_{have}	f_{later}	f_{me}	f_{not}	f_{now}	$f_{unsubscribe}$	f_{won}
Sorry I'll call later														
U can call me now														
U have won call now!!														
Sorry! U can not unsubscribe														

Classwork

	$f_{!}$	$f_{'ll}$	f_{I}	f_{Sorry}	f_{U}	f_{call}	f_{can}	f_{have}	f_{later}	f_{me}	f_{not}	f_{now}	$f_{unsubscribe}$	f_{won}
Sorry I'll call later	0	1	1	1	0	1	0	0	1	0	0	0	0	0
U can call me now	0	0	0	0	1	1	1	0	0	1	0	1	0	0
U have won call now!!	2	0	0	0	1	1	0	1	0	0	0	1	0	1
Sorry! U can not unsubscribe	1	0	0	1	1	0	1	0	0	0	1	0	1	0

What now?

- So, we have learned what features are
- Now, how are we going to use these features to classify a natural language data?
- Let’s see one of the easiest types of classifier– it uses probability

Probability review

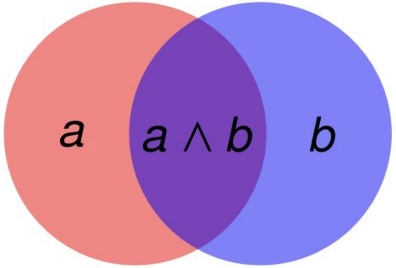
$P(a)$ is our degree of belief in proposition a

- $P(SkyInTucson = sunny) = 0.78$

Probability rules:

$$0 \leq P(a) \leq 1$$

$$\begin{aligned} P(true) &= 1 \\ P(false) &= 0 \\ P(a \lor b) &= P(a) + P(b) - P(a \land b) \end{aligned}$$



Prior Probability

- The unconditional or prior probability of a proposition a is the degree of belief in that proposition given no other information

- P(DieRoll = 5) = 1/6
- P(CardDrawn = A♠) = 1/52
- P(SkyInTucson = sunny) = 286/365

Joint Probability

- The joint probability of propositions a1, . . . , an is the degree of belief in the proposition $a_1 \wedge ... \wedge a_n$

- P(A,♠) = P(A ∧ ♠) = 1/52

Conditional Probability

- The posterior or conditional probability of a proposition a given a proposition b is the degree of belief in a, given that we know only b

- P(Card = A♠|CardSuit = ♠) = 1/13
- P(DieRoll2 = 5|DieRoll1 = 5) =

Relation between Joint and Conditional

Product Rule

- $P(a \land b) = P(a|b)P(b)$ or $P(a|b) = P(a \land b)/P(b)$

- $P(A \land ♠) = 1/52$
- $P(A|♠) = 1/13$
- $P(♠) = 1/4$
- $P(A|♠)P(♠) = 1/13 \cdot 1/4 = 1/52$

Bayes' Rule

$P(b|a) = P(a|b)P(b)/P(a)$

Let's derive this equation.

Purpose of Bayes' rule: Swap the conditioned and conditioning variables

Conditional independence

- Naïve Bayes classifier assumes conditional independence
- Two events A and B are conditionally independent given an event C if

$$P(A \cap B|C)=P(A|C)P(B|C)$$

- A more general version:

$$P(A_1\ A_2\ A_3\ \dots\ A_n|C) = \prod P(A_i|C)$$

Probabilistic classifiers

A probabilistic classifier, h, predicts the category of an object, o, as the most likely category given the object:

$$h(o) = \text{argmax}_{y \in Y} P(y|o)$$

Applying Bayes' rule yields:

$$h(o) = \text{argmax}_{y \in Y} \frac{P(o|y)P(y)}{P(o)}$$

And since $P(o)$ does not depend on y:

$$h(o) = \text{argmax}_{y \in Y} P(o|y)P(y)$$

Applying feature functions:

$$h(o) = \text{argmax}_{y \in Y} P(\left[\begin{array}{cccc} f_1(o) & f_2(o) & \cdots & f_m(o) \end{array}\right] |y) \cdot P(y)$$

Naïve Bayes classifiers

A naïve Bayes classifier is a probabilistic classifier that assumes that the features are independent given the class:

$$P(\begin{bmatrix} f_1(o) & f_2(o) & \cdots & f_m(o) \end{bmatrix} | y) \\ = P(f_1(o)|y) \cdot P(f_2(o)|y) \cdot \ldots \cdot P(f_m(o)|y) \\ = \prod_{i=1}^{m} P(f_i(o)|y)$$

A naïve Bayes classifier thus makes predictions as:

$$h(o) = \operatorname*{argmax}_{y \in Y} P(y) \prod_{i=1}^{m} P(f_i(o)|y)$$

Let’s classify

- Or, in other words, train a classifier on train data and predict the class of a new data
- We will need:
    - Prior probability of a class y; P(y)
    - And conditional probability of each feature f_i $P(f_i|y)$
- Where do we get these probabilities from?
    - From the training data
    - This is the training phase
- Recall from the equation in previous slide:
    - We multiply all these probabilities to get the final probability of a test data
    - This is the test phase

Let's clear this up with an example

$$h(o) = \text{argmax}_{y \in Y} P(y) \prod_{i=1}^{m} P(f_i(o)|y)$$

Training data:	$f_!$	$f_{'ll}$	f_I	f_{Sorry}	f_U	f_{call}	f_{can}	f_{have}	f_{later}	f_{me}	f_{now}	f_{won}
+ *Sorry I'll call later*	0	1	1	1	0	1	0	0	1	0	0	0
+ *U can call me now*	0	0	0	0	1	1	1	0	0	1	1	0
– *U have won call now!!*	2	0	0	0	1	1	0	1	0	0	1	1
Testing data:												
Sorry! U can not unsubscribe	1	0	0	1	1	0	1	0	0	0	0	0

$$P(+|o) \propto P(+)P(f_!|+)P(f_{Sorry}|+)P(f_U|+)P(f_{can}|+)$$
$$P(-|o) \propto P(-)P(f_!|-)P(f_{Sorry}|-)P(f_U|-)P(f_{can}|-)$$

We have faced our first issue!

- Multiplying probability always has this type of issue: if a probability is 0, the entire multiplication becomes 0
- What should we do?
- We use a technique called smoothing

$$P(f_i(o) = k|y) = \left(\frac{1 + \text{count of } i \text{ with } y}{m + \text{count of any with } y}\right)$$

m = number of unique words in the training data

Let's try to do this example again, but now with smoothing

$$h(o) = \text{argmax}_{y \in Y} P(y) \prod_{i=1}^{m} P(f_i(o)|y)$$

$$P(f_i(o) = k|y) = \left(\frac{1 + \text{count of } i \text{ with } y}{m + \text{count of any with } y} \right)$$

Training data:	$f_!$	$f_{'ll}$	f_I	f_{Sorry}	f_U	f_{call}	f_{can}	f_{have}	f_{later}	f_{me}	f_{now}	f_{won}
+ *Sorry I'll call later*	0	1	1	1	0	1	0	0	1	0	0	0
+ *U can call me now*	0	0	0	0	1	1	1	0	0	1	1	0
– *U have won call now!!*	2	0	0	0	1	1	0	1	0	0	1	1
Testing data:												
Sorry! U can not unsubscribe	1	0	0	1	1	0	1	0	0	0	0	0

$$P(+|o) \propto P(+)P(f_!|+)P(f_{Sorry}|+)P(f_U|+)P(f_{can}|+)$$

$$P(-|o) \propto P(-)P(f_!|-)P(f_{Sorry}|-)P(f_U|-)P(f_{can}|-)$$

After the calculation

$$P(+|o) \propto \frac{2}{3} \cdot \frac{1}{22} \cdot \frac{2}{22} \cdot \frac{2}{22} \cdot \frac{2}{22} \approx 0.000023$$

$$P(-|o) \propto \frac{1}{3} \cdot \frac{3}{19} \cdot \frac{1}{19} \cdot \frac{2}{19} \cdot \frac{1}{19} \approx 0.000015$$

$$\Rightarrow \quad h(o) = +$$

Refining features

- Binarization
- Binary multinomial naïve Bayes assumes that what matters is whether or not a word occurs in a document, not its frequency in the document.
- Equivalent to removing duplicate words in each document.

$$
\begin{array}{cccc}
f_{!} & f_{'ll} & f_{I} & f_{Sorry} \\
\hline
0 & 1 & 1 & 3 \\
0 & 0 & 0 & 0 \\
2 & 0 & 0 & 0 \\
1 & 0 & 0 & 1 \\
\hline
\end{array}
\Rightarrow
\begin{array}{cccc}
f_{!} & f_{'ll} & f_{I} & f_{Sorry} \\
\hline
0 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 \\
1 & 0 & 0 & 1 \\
\hline
\end{array}
$$

Refining features

- N-gram features
- Instead of one word at a time, consider multiple words
- Bag-of-words assumption is sometimes too simplistic:
  - good vs. not good
  - like vs. didn't like
- A bag-of-n-grams feature representation has a vocabulary that includes phrases (up to) n tokens long.
  - Example: $f_{\text{great film}}$ (great scenes great film) = 1
  - This is bag-of-bigrams feature
- bag-of-words = bag-of-1-grams (a.k.a. unigrams)
- Consequences:
  - Vocabulary size grows quickly as n increases
  - Small counts; most n-grams never seen for large n
- For word n-grams, typically $n \leq 3$
- For character n-grams, typically $n \leq 10$

Refining features

- Lexicon features
- Sometimes there isn't enough training data.
    - Example: sentiment training data that doesn't include wretched, dreadful, genial, etc.
- A lexicon is a list of words (not sentences or documents) that have been annotated for a task.
    - Example: the LIWC lexicon labels words for the categories Positive emotion, Family etc.
- Lexicons are typically used to produce count features.
    - Example: f_{posemo} (great scenes beautiful film) = 2 if
    - great and beautiful are tagged as posemo in the lexicon

Rule-based features

- There are no limits on the algorithm that a feature function may apply.
- Examples:
  - How many times does [$][\d,]{7} match?
  - What proportion of first line’s characters are capitals?
  - What is the ratio of text to image area (in HTML)?

Worked Example: Logistic Regression

Use LR to classify the following reviews using BOW. Given,

Vocabulary = {very, good, movie, not, bad}

Word2Index = {very = 1, good = 2, movie = 3, not = 4, bad = 5}

Weights, [w1,w2,w3,w4,w5] = [0.3 , 1.5 , 1 , -1.5 , -2]   Bias, b = -0.5

Class = [0 = Not Spam, 1 = Spam]

Reviews:

1. not that bad
2. very good

Solution

Vocabulary = {very, good, movie, not, bad}
Word2Index = {very = 1, good = 2, movie = 3, not = 4, bad = 5}
Weights, [w1,w2,w3,w4,w5] = [0.3 , 1.5 , 1 , -1.5 , -2]   Bias, b = -0.5

BOW vector of Review 1: not that bad = [x1,x2,x3,x4,x5] = [0,0,0,1,1]
We know,

$$z = w1*x1 + w2*x2 + w3*x3 + w4*x4 + w5*x5 + b$$
$$= 0.3 * 0 + 1.5 * 0 + 1 * 0 + (-1.5) * 1 + (-2) * 1 - 0.5$$
$$= -4$$
$$\hat{y} = \sigma(z) = 1/(1-\exp(-(-4))) = 0.018 \text{ which is less than } 0.5$$

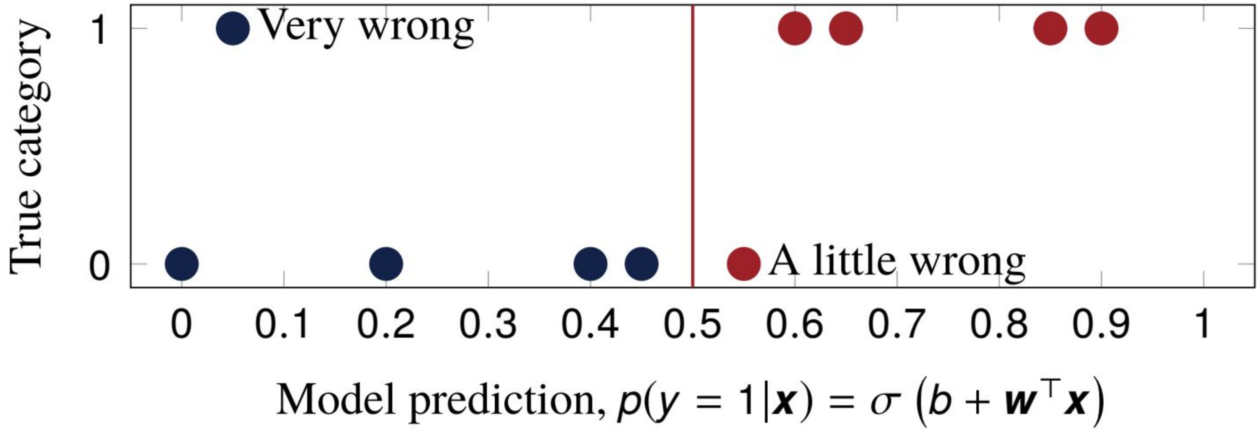
So predicted class = 0 = negative
Do the same for review 2:

Probabilistic learners are simple

- In everyday life, the learners we use make mistakes and then learn from mistakes
    - Let’s say we want to draw a line to separate spans vs. not spams (linear regression classifier)
    - The line splits the spams and not spams into two regions
    - If a spam falls in the non-spam region, the classifier made a mistake
    - This line is called a “decision boundary”
- How do we do that?
    - We can penalize the model for making mistakes
    - How can we penalize? Should we penalize the same amount for every mistake it makes?
    - Enter the idea of *Loss*

Loss

How "wrong" is this classifier?



Classification errors: $Cost_{0/1}(\boldsymbol{w}, b; \boldsymbol{X}, \boldsymbol{y}) = \dfrac{2}{10}$

But not all errors are equal

- Errors near the decision boundary are "less bad"

Cross entropy loss

Our estimated $p(y = 1|\boldsymbol{x}) = \sigma\left(b + \boldsymbol{w}^\top \boldsymbol{x}\right) = \hat{y}$ is good if:

- $\hat{y}$ is high when the true label is $y = 1$
- $\hat{y}$ is low when the true label is $y = 0$

So a good classifier would maximize:

$$p(y|\boldsymbol{x}) = \hat{y}^y (1 - \hat{y})^{(1-y)}$$

Or, equivalently, minimize:

$$\begin{aligned}
L_{CE}(\boldsymbol{w}, b; \boldsymbol{x}, y) &= -\log p(y|\boldsymbol{x}) \\
&= -\log\left(\hat{y}^y (1 - \hat{y})^{(1-y)}\right) \\
&= -\left(y \log \hat{y} + (1 - y) \log(1 - \hat{y})\right) \\
&= -\Big(y \log \sigma\left(b + \boldsymbol{w}^\top \boldsymbol{x}\right) + \\
&\qquad (1 - y) \log\left(1 - \sigma\left(b + \boldsymbol{w}^\top \boldsymbol{x}\right)\right)\Big)
\end{aligned}$$

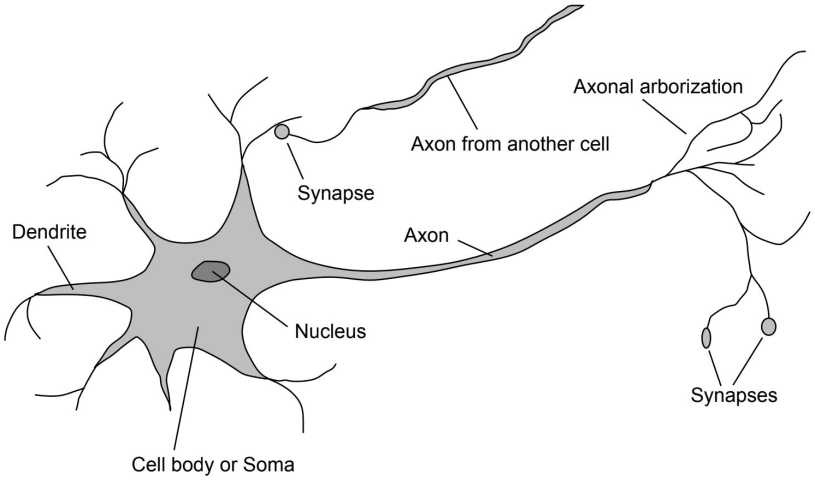
Cross entropy loss

- This cross-entropy loss, $L_{CE}(\mathbf{w}, b; \mathbf{x}, y)$, measures how bad **w** and b are on the single example ($\mathbf{x}$, y).
- An ML model runs through each of the training examples and tries to reduce this cross entropy loss as it goes along– and thus learns the pattern in the examples
- Sometimes, a model goes through all training examples multiple times to reduce the loss further
    - Each of these run is called an epoch

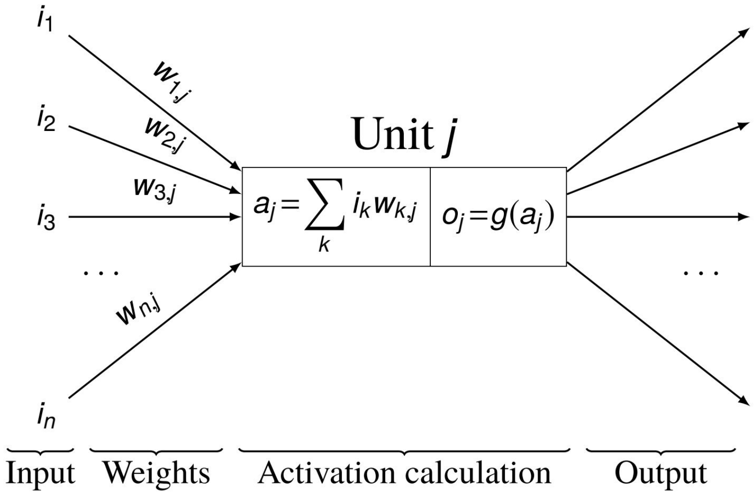
Lecture 3.1: Neural Nets

Neural Networks

Neuron in a human brain

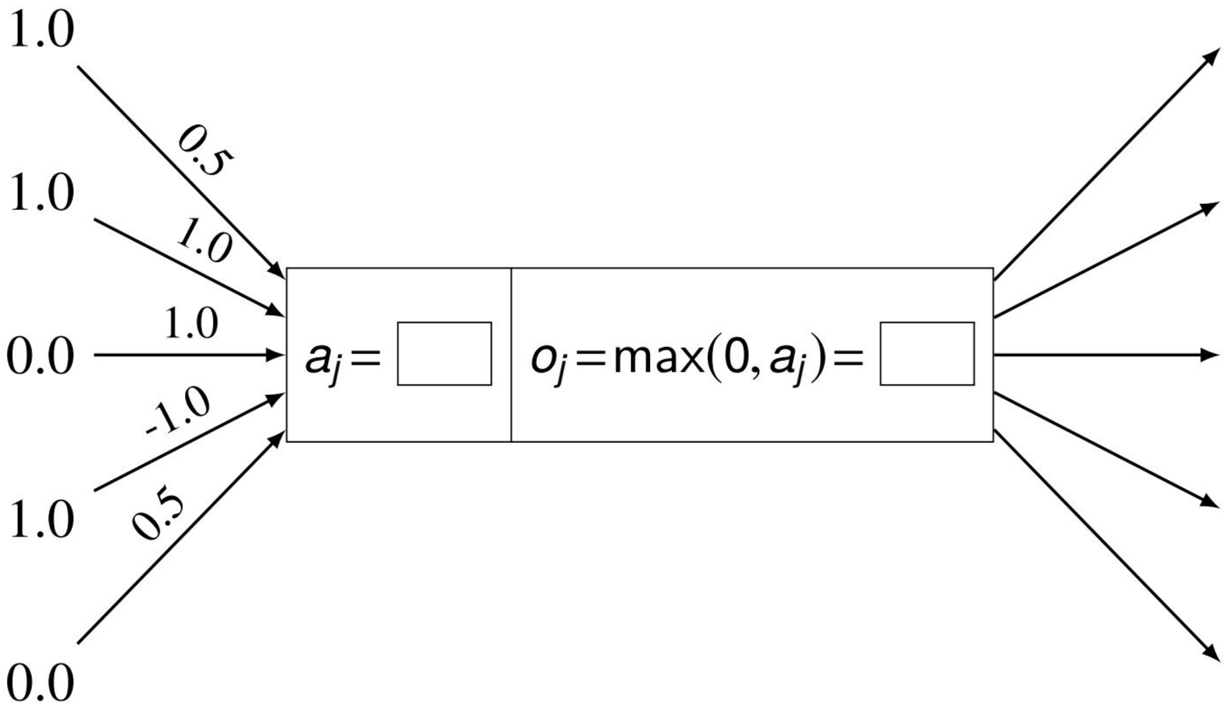


Neuron in an ML model



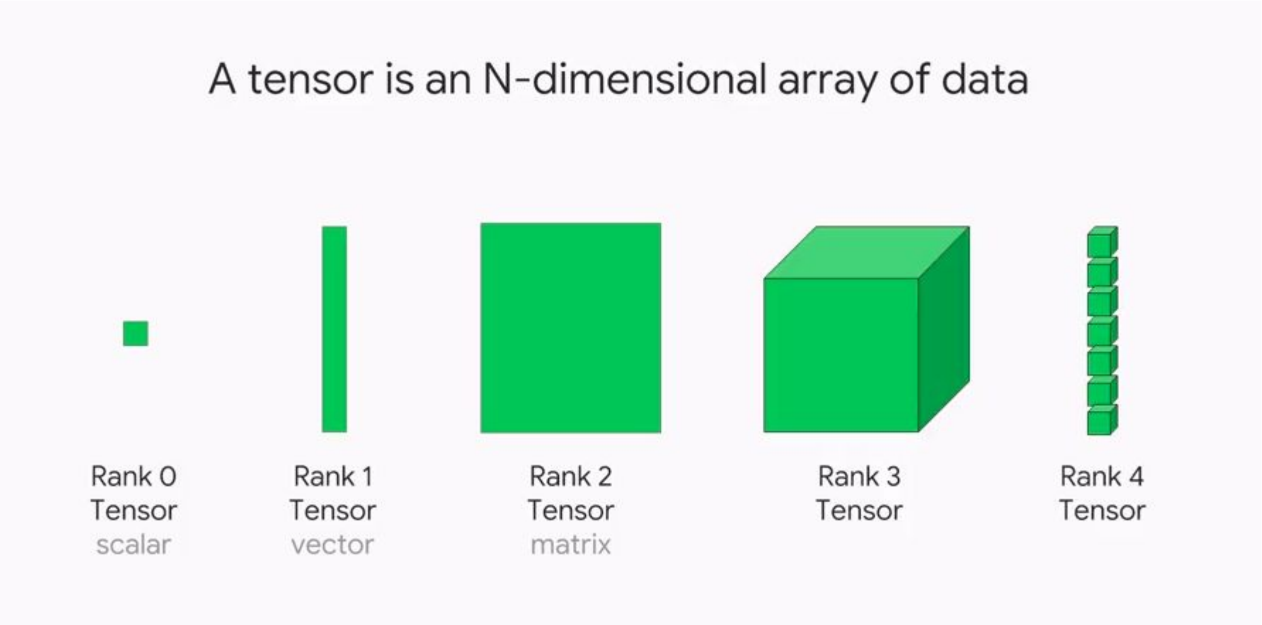
Class work

Calculate the output of this unit:



Unit calculations as tensor arithmetic

What is a tensor?



Unit calculations as tensor arithmetic

Summation for a single unit:

$$o_j = g\left(\sum_k i_k w_{k,j}\right)$$

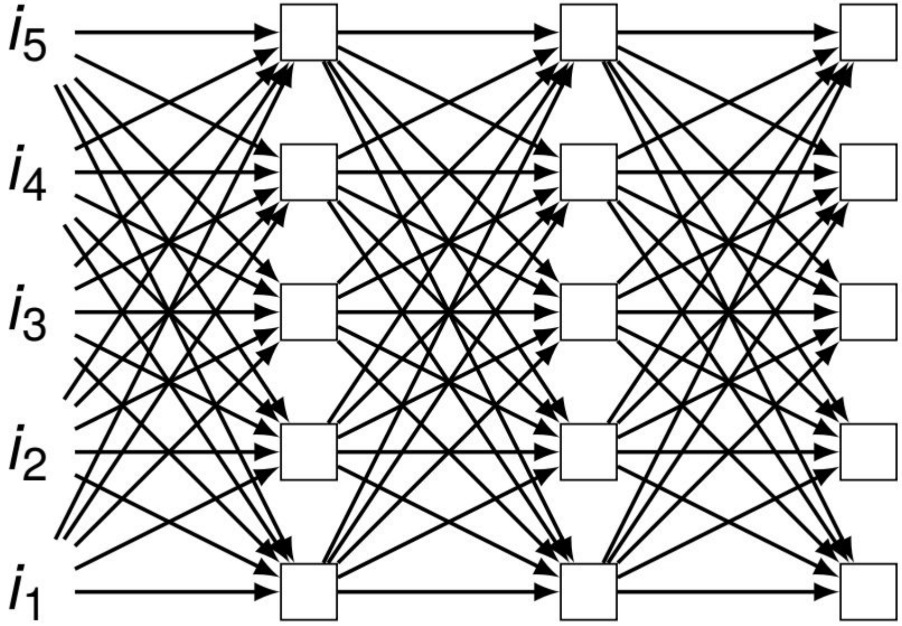
Vector arithmetic for a single unit:

$$o_j = g\left(\begin{bmatrix} i_1 & i_2 & \dots & i_n \end{bmatrix} \begin{bmatrix} \boldsymbol{w}_{1,j} \\ \boldsymbol{w}_{2,j} \\ \dots \\ \boldsymbol{w}_{n,j} \end{bmatrix}\right) = g\left(\boldsymbol{i} \times \boldsymbol{w}_{*,j}\right)$$

Matrix arithmetic for multiple units:

$$\boldsymbol{o} = g\left(\boldsymbol{i} \times \boldsymbol{W}\right)$$

A feedforward network as composition



$$\boldsymbol{o} = g\left(g\left(g\left(\boldsymbol{i} \times \boldsymbol{W}^1\right) \times \boldsymbol{W}^2\right) \times \boldsymbol{W}^3\right)$$

Activation functions

- Why do we need activation functions?
- What types of activation functions do we have?

Learning the XOR

$$\boldsymbol{X} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \qquad \boldsymbol{y} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

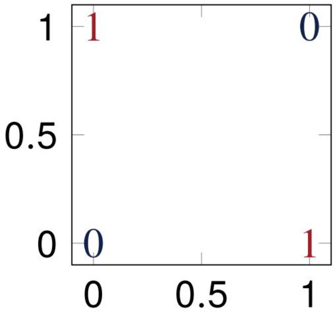
Can you solve it with linear regression?

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{W} + \boldsymbol{b}$$

$$\begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \boldsymbol{W} + \boldsymbol{b}$$

$$\begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} ? \\ ? \end{bmatrix} + ?$$

No such weights exist!



Solving XOR with NNs

$$f(\boldsymbol{X};\boldsymbol{W},\boldsymbol{c},\boldsymbol{w},b) = \max\left(0,\boldsymbol{X}\boldsymbol{W}+\boldsymbol{c}\right)\boldsymbol{w}+b$$

$$\begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \max\left(0, \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \boldsymbol{W} + \boldsymbol{c}\right)\boldsymbol{w} + b$$

$$\begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \max\left(0, \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} ? & ? \\ ? & ? \end{bmatrix} + \begin{bmatrix} ? & ? \end{bmatrix}\right)\begin{bmatrix} ? \\ ? \end{bmatrix} + ?$$

Solving XOR with NNs

$$\begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \max\left(0, \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -1 \end{bmatrix}\right) \begin{bmatrix} 1 \\ -2 \end{bmatrix} + 0$$

$$\begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} + 0$$

Common activation functions in hidden units

We have: affine transformation of input x, followed by nonlinear activation function

$$\boldsymbol{h} = g(\boldsymbol{W}^\top \boldsymbol{x} + \boldsymbol{b})$$

g could be just about anything! It can be linear, but a linear function is not preferred. Why?

Considerations:

- What specific behavior is needed?
- How will the gradients behave?

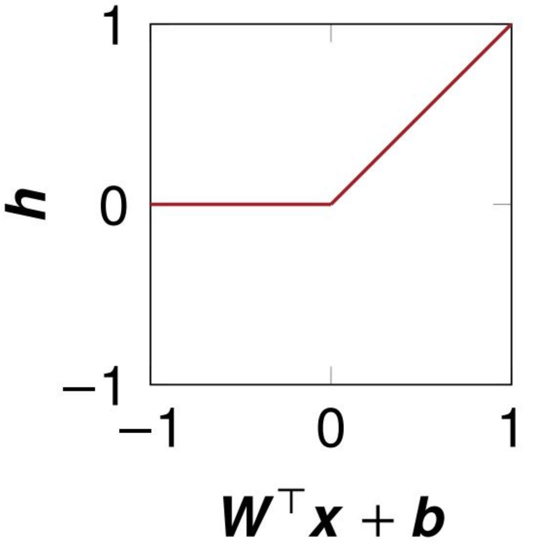
Why linear functions are not preferred?

Because of the considerations.

- We need complex mappings between the inputs and the outputs
- All linear layers will translate the input linearly to output– that is, multiple linear transformation is basically one giant linear transformation
- Cannot use backpropagation as the derivative is constant

ReLU

$$\boldsymbol{h} = \max(0, \boldsymbol{W}^\top \boldsymbol{x} + \boldsymbol{b})$$



Behavior?

- Active only when input is positive

Gradients?

- 1 when positive
- 0 when negative

ReLU is non-differentiable

ReLU at $z = 0$:

- left derivative = 0
- right derivative = 1

So ReLU is not differentiable at $z = 0$!

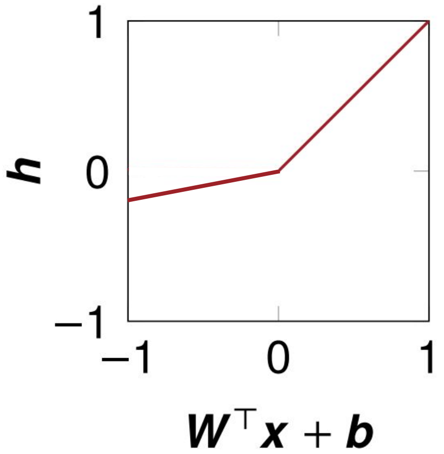
A few non-differentiable points are not a problem:

- Training rarely reaches a point with gradient 0 anyway
- Software simply returns either left or right derivative

Leaky ReLU

$$f(x) = max\ (0.1x, x)$$

$$\boldsymbol{h} = \max(0, \boldsymbol{W}^\top \boldsymbol{x} + \boldsymbol{b}) + 0.01 \min(0, \boldsymbol{W}^\top \boldsymbol{x} + \boldsymbol{b})$$



ReLU has a “dead neuron” problem!

Behavior?

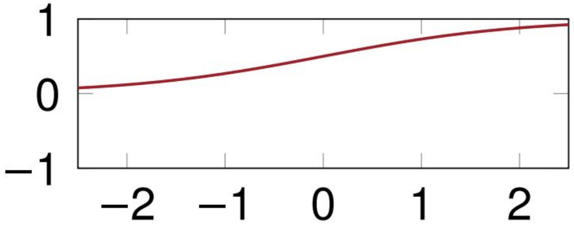
- Strong positive activation when positive
- Very weak negative activation when negative

Gradients?

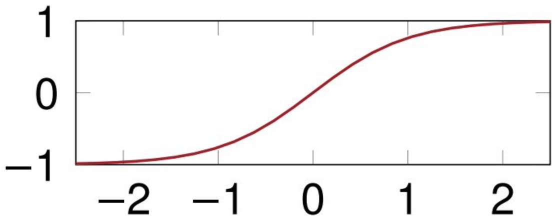
- 1 when positive
- 0.1 when negative

Sigmoid and tanh

Sigmoid: $f(x) = \dfrac{1}{1+e^{-x}}$



Tanh: $f(x) = \dfrac{(e^{x} - e^{-x})}{(e^{x} + e^{-x})}$



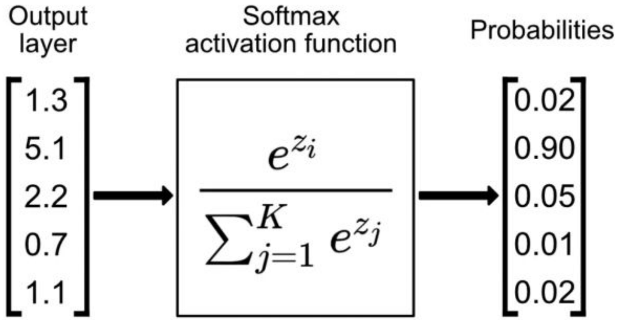
Behavior?

- Sigmoid: 0/1 switch
- Tanh: -1/+1 switch

Gradients?

- Saturate across most of their domain
- Tanh optimizes slightly better since it is similar to the identity function near 0

Softmax



Softmax converts raw model outputs into probabilities across multiple classes.

It ensures all output values are **between 0 and 1** and **sum to 1.**

Higher input scores receive higher probabilities, making it suitable for multi-class classification.

Hyperparameters

- Network parameters are the ones that are being learned throughout the training process (e.g. weights)
- There are parameters that we can control to facilitate this learning: they are called hyperparameters
- Activation function is one such hyperparameter
- We have others …

Optimizers

- Algorithms used to update the learnable parameters in order to reduce loss
- Common optimizers:
  - Gradient descent
  - Stochastic gradient descent
  - Mini-batch gradient descent
  - Momentum
  - Adagrad
  - RMSProp
  - AdaDelta
  - Adaptive moment estimation
- GD methods maintain a single learning rate (with/without decay for all parameters and are known as first order optimizers (works with the first order derivative)
- Adaptive methods like Adagrad provide learning rates for each parameter, thus improving the learnability, but are computationally expensive
- RMSProp not only provides LR for each parameter, it adapts based on the mean of recent magnitudes of the gradients for the weight → first order. AdaDelta is similar but works with squared gradients (second order)

Adaptive moment estimation

- Popularly known as Adam (not ADAM)
- Came out of OpenAI and University of Toronto
- Most likely the highest cited paper in recent history
- Why is it so popular?
  - Straightforward to implement
  - Little memory requirements
  - Well suited for problems that are large in terms of data and/or parameters
  - Needs little to no manual tuning

Adam

- Adam works with momentums of first and **second** order
- Instead of adapting the parameter learning rates based on the average first moment (the mean), m_t as in RMSProp, Adam also makes use of the average of the second moments of the gradients (the uncentered variance) v_t

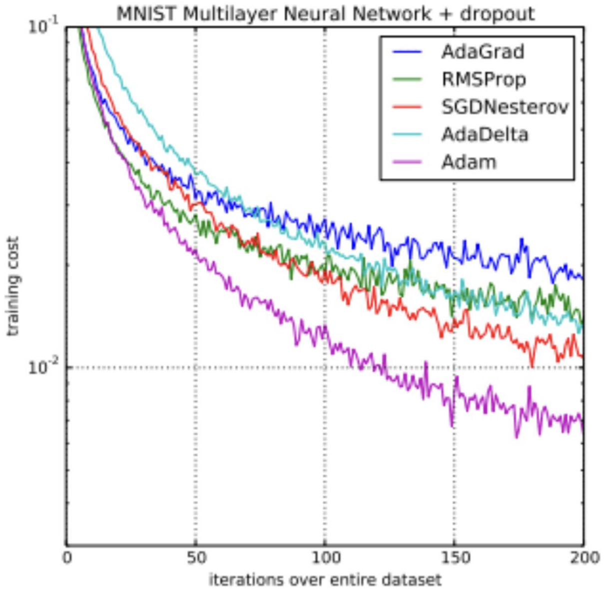
$$\hat{m}_t = \frac{m_t}{1-\beta_1^t}.$$

$$\hat{v}_t = \frac{v_t}{1-\beta_2^t}.$$

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon}\hat{m}_t.$$

- The values for β1 is 0.9 , 0.999 for β2 and epsilon is an extremely small number to avoid zero division

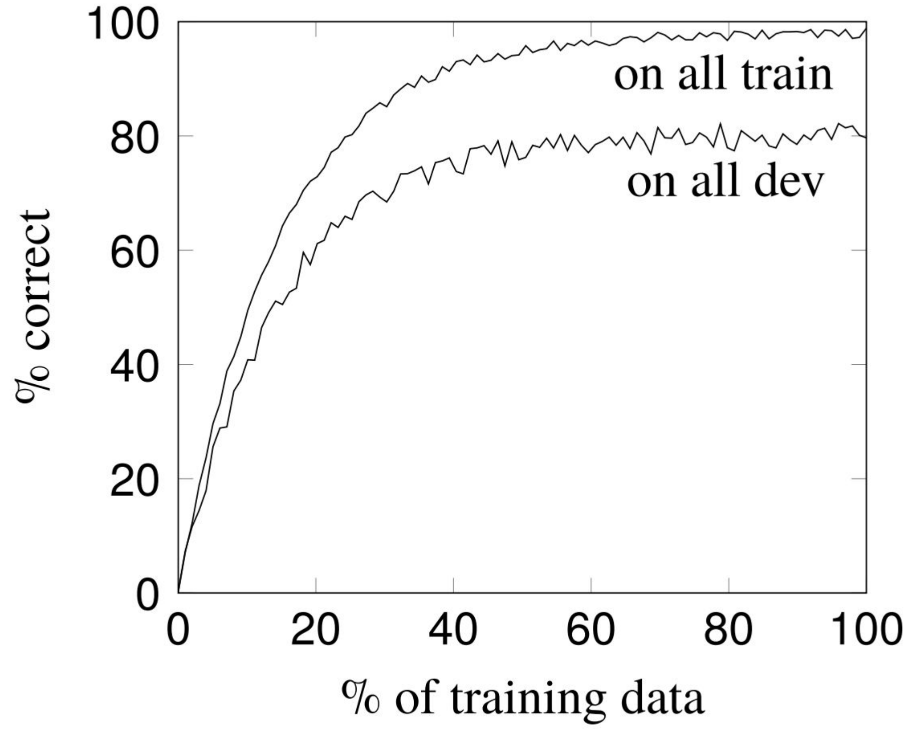
Adam’s popularity



Sebastian Ruder: “... RMSprop, Adadelta, and Adam are very similar algorithms that do well in similar circumstances. ….. Adam might be the best overall choice.”
Andrej Karpathy: “In practice Adam is currently recommended as the default algorithm to use, and often works slightly better than RMSProp.”

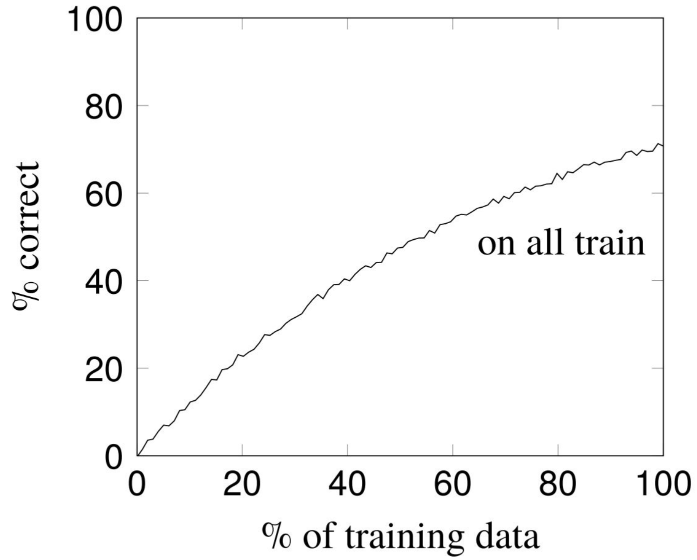
Underfitting and Overfitting

Learning Curve

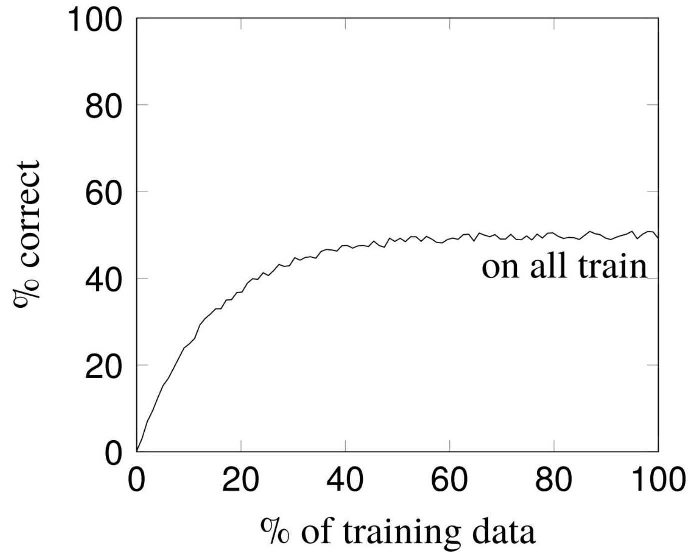


Underfitting

Insufficient training data



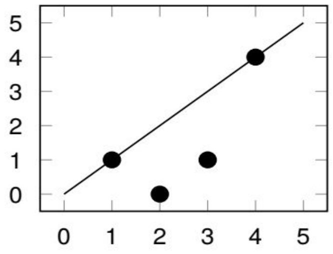
Model is too simple



Addressing underfitting

x_1	$f(x)$	$h(x)$
1	1	1
2	0	2
3	1	3
4	4	4

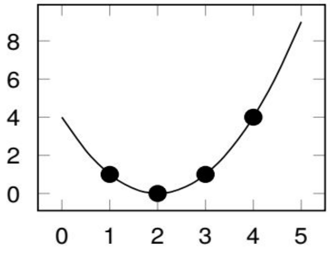
$$h(x) = x_1$$



More complex model

x_1	$f(x)$	$h(x)$
1	1	1
2	0	0
3	1	1
4	4	4

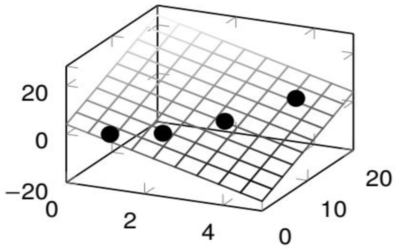
$$h(x) = x_1^2 - 4x_1 + 4$$



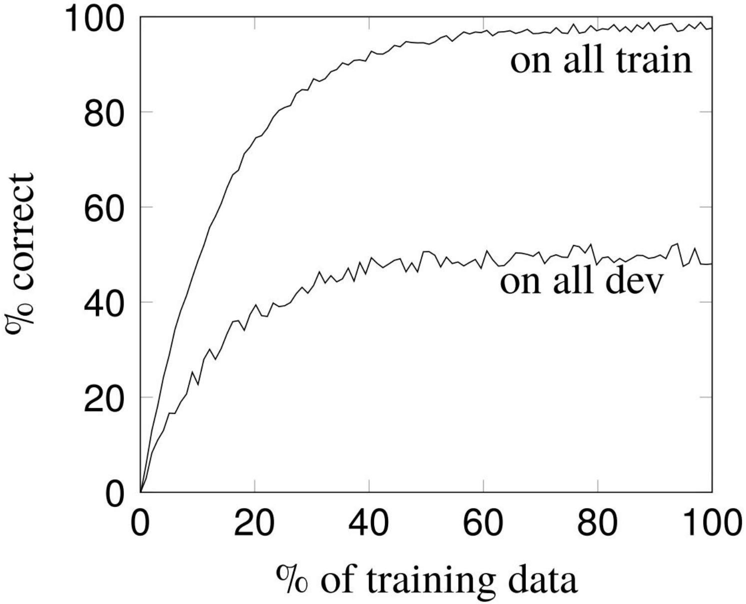
More features

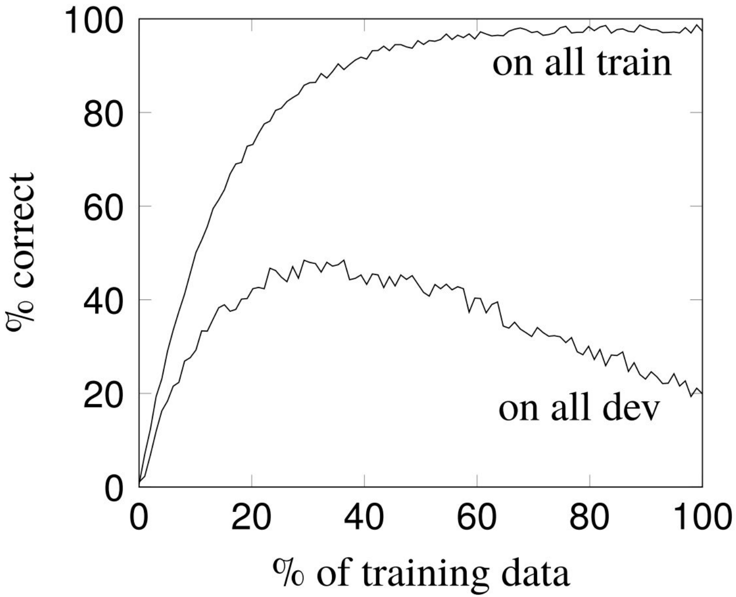
x_1	x_2	$f(x)$	$h(x)$
1	1	1	1
2	4	0	0
3	9	1	1
4	16	4	4

$$h(x) = -4x_1 + x_2 + 4$$



Overfitting

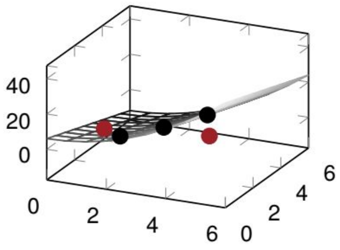




Addressing overfitting

x_1	x_2	$f(x)$	$h(x)$
2	1	7	7
3	2	9	9
4	3	13	13
1	2	3	1
5	1	15	28

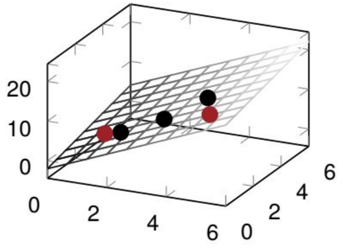
$$h(x) = x_1^2 - 3x_2 + 6$$



Simpler model

x_1	x_2	$f(x)$	$h(x)$
2	1	7	5
3	2	9	9
4	3	13	13
1	2	3	3
5	1	15	14

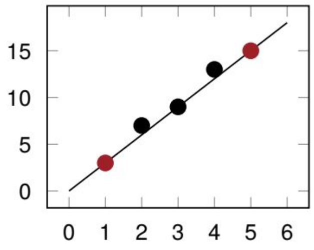
$$h(x) = 3x_1 + x_2 - 2$$



Fewer features

x_1	$f(x)$	$h(x)$
2	7	6
3	9	9
4	13	12
1	3	3
5	15	15

$$h(x) = 3x_1$$

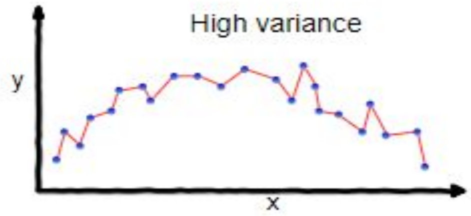


Regularization

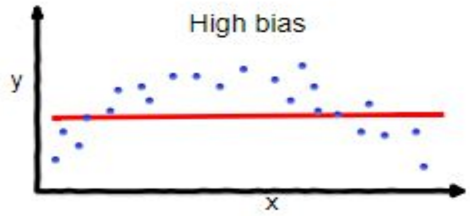
- A set of strategies used in Machine Learning to reduce the generalization error
- Why?
- Bias-variance tradeoff
- Bias: error from wrong assumptions in the learning algorithm
  - High bias can cause an algorithm to miss the relevant relations between features and target outputs → Observations don't matter
  - Underfitting
- Variance: error from sensitivity to small fluctuations in the training set
  - High variance may result in modeling the random noise in the training data → focusing too much on observations
  - Overfitting

Regularization

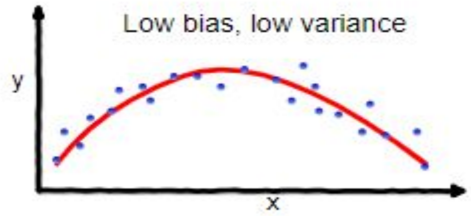
- A good model needs to balance bias and variance
- Regularization helps us do that



overfitting



underfitting



Good balance

Regularization techniques

- Introduce regularization term to the loss function
    - Introduces a small amount of bias to counter variance → reduce overfitting
    - Loss function: negative log likelihood or binary cross entropy
    - Most common terms:
        - L2 regularizer: Ridge regression → adds the “squared magnitude” of the coefficient as the penalty term to the loss function
        - L1 regularizer: Lasso regression → adds the “absolute value of magnitude” of the coefficient as a penalty term to the loss function
        - Uses λ that controls the sensitivity of`the model to the input → less sensitive, less likely to overfit
        - L2 focuses on larger weights, so higher λ means L2 penalizes higher value weights more than lower ones → no feature essentially goes away
        - L1 has equal focus, so, higher λ → low weight features go away → feature selection

Regularization techniques

Loss = Original Loss + (λ* Regularization term)

L2: Ridge Regression

Penalty = $\lambda^* \sum w_i^2$

- Penalize the large weight more strongly
- Keeps all features, but shrinks their influence
- High λ → smaller weights → less complex model → prevent overfitting

L1: Lasso Regression

Penalty = $\lambda^* \sum |w_i|$

- Adds the absolute weight to the loss
- High λ → more weights become 0
- Zero weights means the feature does not affect the model
- less complex model → prevent overfitting

Regularization techniques

- Dropout: Drops out (ignore) a layer’s output with a probability *p*
    - Choice of p depends on the architecture
- Early stopping: Stops when performance gets saturated
    - Stops model from being overfit
    - Returns the best current model
- Data augmentation
    - Introduce new training data with variation
    - Injects noise

Other common hyperparameters

- Learning rate: how quickly a network updates its parameters
    - Low learning rate slows down the learning process but converges smoothly
    - Larger learning rate speeds up the learning but may not converge
    - Decaying learning rate is preferred
- Epochs: How many times the entire training dataset has passed through the model
- Batch size: (Mini) batch size refers to a subset of the training data. Weights are updated after each mini batch training
    - Small mini batch → too many updates
    - Large mini batch → too few updates, too many epochs to converge

ML best practices

- Data splitting
- Underfitting and overfitting
- Performance metrics
- Statistical significance

Data splitting

- Train classifier on data E_{train}
- Test classifier on E_{test}

```
>>> from sklearn.feature_extraction.text import TfidfVectorizer
>>> from sklearn.model_selection import train_test_split
>>> corpus = [
... 'This is the first document.',
... 'This document is the second document.',
... 'And this is the third one.',
... 'Is this the first document?',
... ]
>>> vectorizer = TfidfVectorizer()
>>> X = vectorizer.fit_transform(corpus)
>>> y = [1, 0, 0, 1]
>>> X_train, X_test, y_train, y_test = train_test_split(
... corpus, y, test_size=0.25, random_state=42)
```

Data splitting

- Never peak into the test data!
  - Not even accidentally

Train:

x_1	x_2	$f(x)$
1	a	*true*
0	b	*false*
0	c	*false*
0	b	*false*

Test:

x_1	x_2	$f(x)$
1	a	*true*
0	b	*false*
0	c	*false*
1	c	*true*

$$h_1(x) = \begin{cases} \mathit{true} & \text{if } x_1 = 1 \\ \mathit{false} & \text{otherwise} \end{cases} \qquad h_2(x) = \begin{cases} \mathit{true} & \text{if } x_2 = a \\ \mathit{false} & \text{otherwise} \end{cases}$$

- What are the accuracies of these models?
- How do you improve them?

Data splitting: improving a model

- Bad idea
  - Train on the train data
  - Evaluate on test data
  - Go back to the training phase and tune parameters
- What will it do?
- Better idea
  - Split data into three parts
    - Train on the major chunk of the data (E_{train})
    - Tune on a very small piece of data (E_{val})
    - Test on another small chunk of data (E_{test})

Performance metrics

- Accuracy is a bad performance metrics
  - the distribution of categories is unbalanced, or
  - you care about one category more than the others
  - Example: detecting spam in text messages
  - Example: diagnosing depression from tweets
- What should we do?
- Use other metrics: precision, recall, F-1 score

Confusion matrix

	Predicted 0	Predicted 1
Actual 0	TN	FP
Actual 1	FN	TP

Precision, recall and F-1 score

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

Precision = ratio between true + (or -) and predicted + (or -)

Recall = ration between true (or -) and actual + (or -)

F-1 score: Harmonic mean of precision and recall

There are other task-specific metrics as well:

- BLEU: machine translation
- WER: ASR
- Cosine similarity: semantic similarity
- Euclidean distance: spell checking
- F-latency: early detection

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Statistical Significance

- On some test set T, classifier A gets .992 F1, B gets .991 F1.
  - Is A actually better than B?
- Competing hypotheses:
  - A is generally better than B
  - A was better than B by chance on T, but would not be better than B in general, i.e., the null hypothesis (H_0)
  - We would like to reject the null hypothesis if we want to establish that A is actually better than B
- How to do this? One way can be:
  - We can do some sort of statistical significance test (Student's t-test) using multiple samples
  - If the test value is less than than a preset critical value (p-value), we reject the null hypothesis
  - How to get the samples? Many ways (e.g. cross-validation)